Communications and Networks

Passband Transmission

Part I: Baseband Equivalent Modeling and Modulation Schemes

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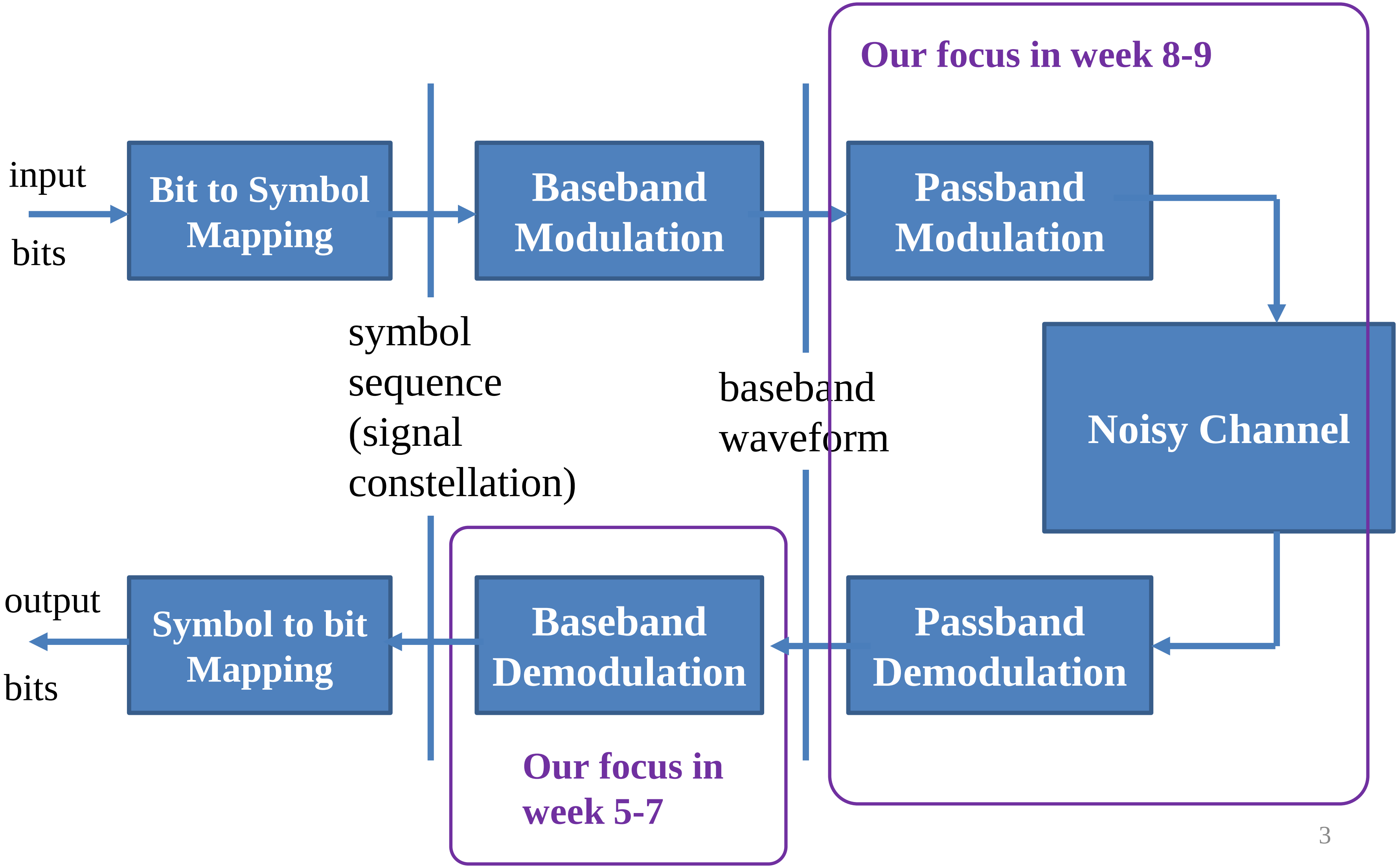
About the Mid-term

- Planned to be on Apr. 25 (in-class)
- Covering the classes till Apr. 18
- Will be open (you can bring any materials in paper, no electronic devices)
- Online access method (ONLY for those NOT on campus) will be announced

Modulation: Overview



Passband Transmission

- Passband transmission
  - Limited/narrow bandwidth channel
  - US DTV: 50-56 MHz
  - Mobile: 900 MHz or 1800 MHz carrier frequency, with 30 kHz or 300 kHz bandwidth
  - Satellite: 12 GHz or 17 GHz carrier frequency, with 26MHz bandwidth
- Passband modulation
  - Modulate the signal onto a carrier (sinusoid) with $\omega_c = 2\pi f_c$
  - Switch the **amplitude, phase,** or frequency of the carrier
  - How to get the passband signal from baseband waveform $s(t) = \sum_k a_k g_k(t)$?

Outline

Part I:

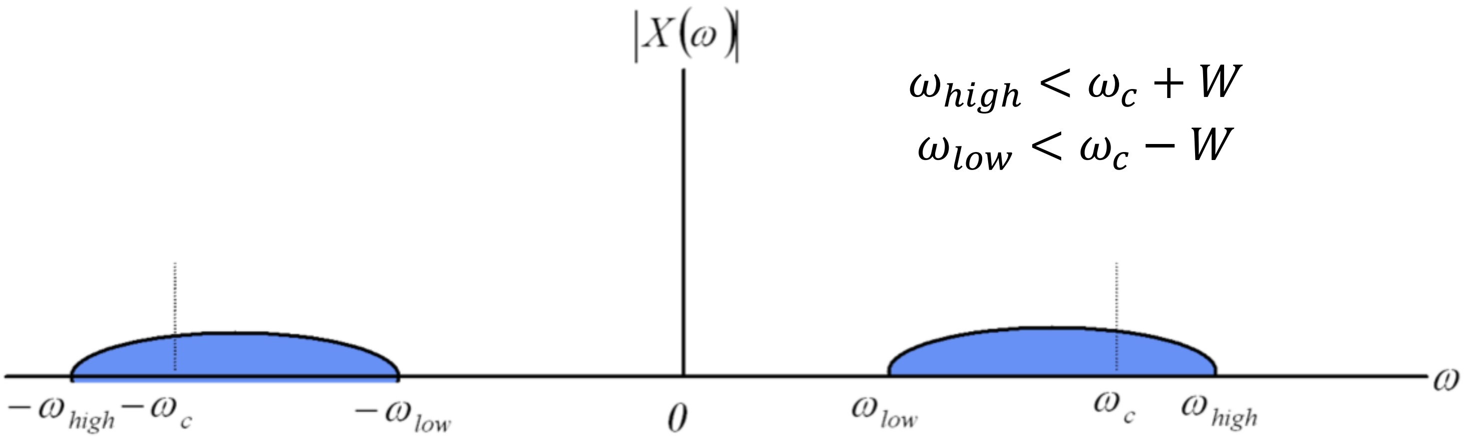
- Baseband equivalent modeling (no noise)
- Passband modulation schemes

Part II:

- Baseband equivalent AWGN channel
- Detection and error analysis

Passband Representations

- Passband signal
  - Real-valued signal $x(t)$ is a passband signal when its nonzero Fourier transform is near ω_c



$$\omega_{high} < \omega_c + W$$
$$\omega_{low} < \omega_c - W$$

  - No DC content.

Passband Representations

- Carrier-Modulated Signal

$$x(t) = A(t)\cos[\omega_c t + \phi(t)]$$

  - $A(t)$: amplitude
  - $\phi(t)$: phase
  - $\omega_c = 2\pi f_c$: carrier frequency

Passband Representations

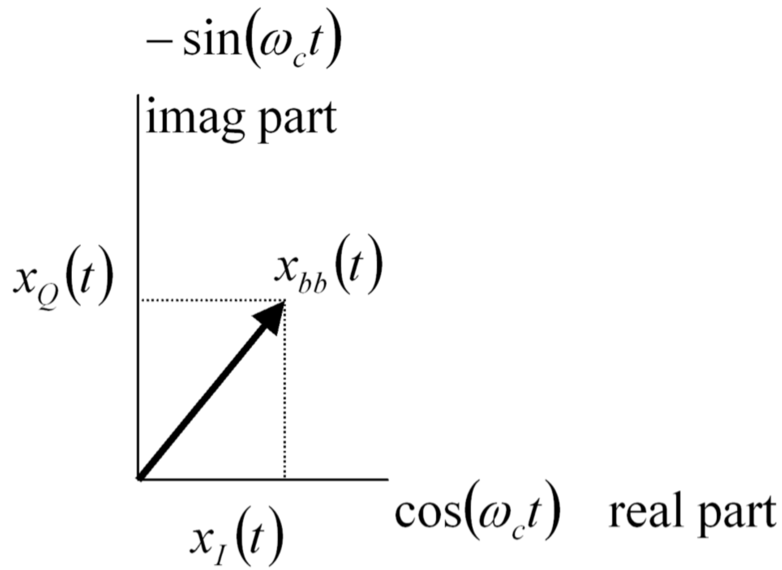
- Quadrature Decomposition

$$x(t) = x_I(t)\cos(\omega_c t) - x_Q(t)\sin(\omega_c t)$$

  - Inphase component: $x_I(t) = A(t)\cos(\phi(t))$
  - Quadrature component: $x_Q(t) = A(t)\sin(\phi(t))$

- Relationships

  - $A(t) = \sqrt{x_I^2(t) + x_Q^2(t)}$
  - $\phi(t) = \tan^{-1}\left[\frac{x_Q(t)}{x_I(t)}\right]$



Baseband Representations

- Goal: reuse our knowledge from baseband transmissions
- Baseband-Equivalent Complex Signal:
  - $x_{bb}(t) \triangleq x_I(t) + jx_Q(t)$
- Analytic-Equivalent Signal
  - $x_A(t) \triangleq x_{bb}(t)e^{j\omega_c t}$
  - $x(t) = \Re\{x_A(t)\} = \Re\{x_{bb}(t)e^{j\omega_c t}\}$

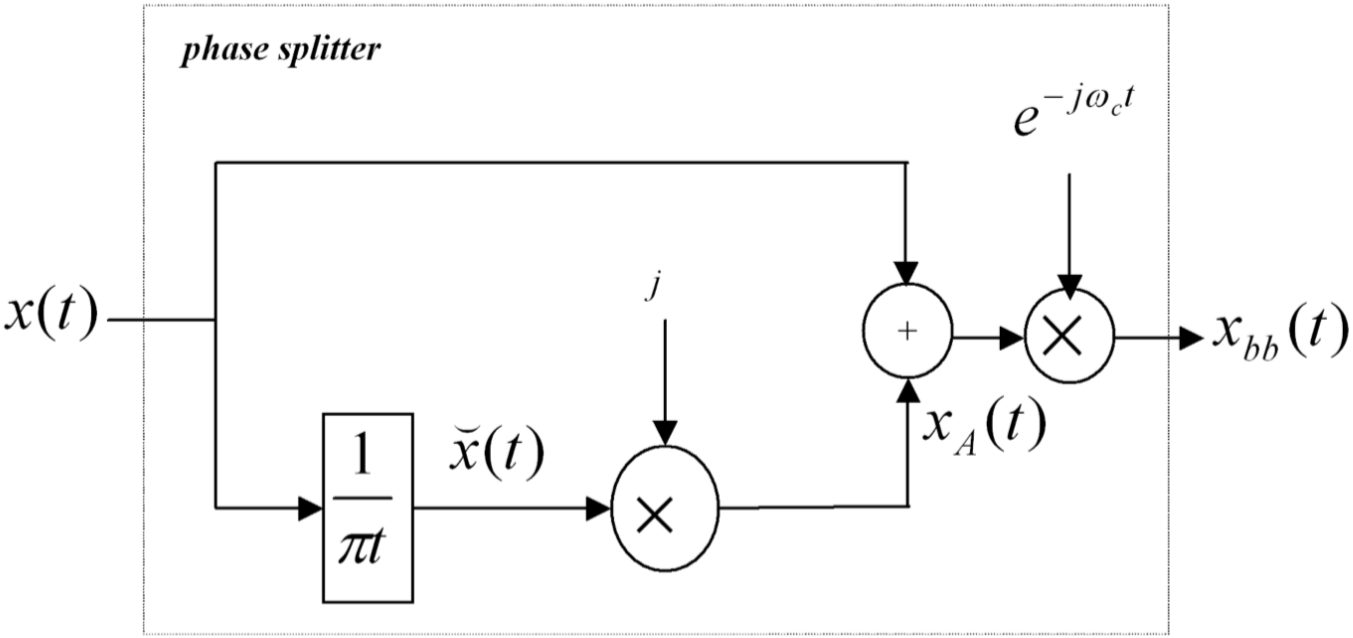
Original passband signal is the real part of the equivalent *rotating* complex baseband representation.

1. Determine passband signal from baseband signal.

Analytic Signal Composition

- Hilbert transform

$$\breve{x}(t) = \Im\{x_A(t)\},\ x_A(t) = x(t) + j\breve{x}(t)$$



2. Determine baseband signal from passband signal.

Representations Summary

- Equivalent forms summary:

$$x(t) = A(t)\cos[\omega_c t + \phi(t)]$$

  - Magnitude, phase $A(t), \phi(t)$

$$x(t) = x_I(t)\cos(\omega_c t) - x_Q(t)\sin(\omega_c t)$$

  - Inphase $x_I(t)$, quadrature, $x_Q(t)$

$$x(t) = \Re\{x_A(t)\}$$

  - Analytic $x_A(t)$

$$x(t) = \Re\{x_{bb}(t)e^{j\omega_c t}\}$$

  - Complex baseband $x_{bb}(t)$

Frequency Spectrum Analysis

- Analytic and baseband signal spectrum
  - $x_A(t) = x(t) + j\check{x}(t)$ — Take Fourier transform
  - $X_A(\omega) = [1 + \operatorname{sgn}(\omega)]X(\omega)$ — $\Re\{X(\omega)\}$ is even and $\Im\{X(\omega)\}$ is odd.

$$= \begin{cases} 2X(\omega) & \omega > 0 \\ X(0) = 0 & \omega = 0 \\ 0 & \omega < 0 \end{cases}$$

Only contains positive frequencies.

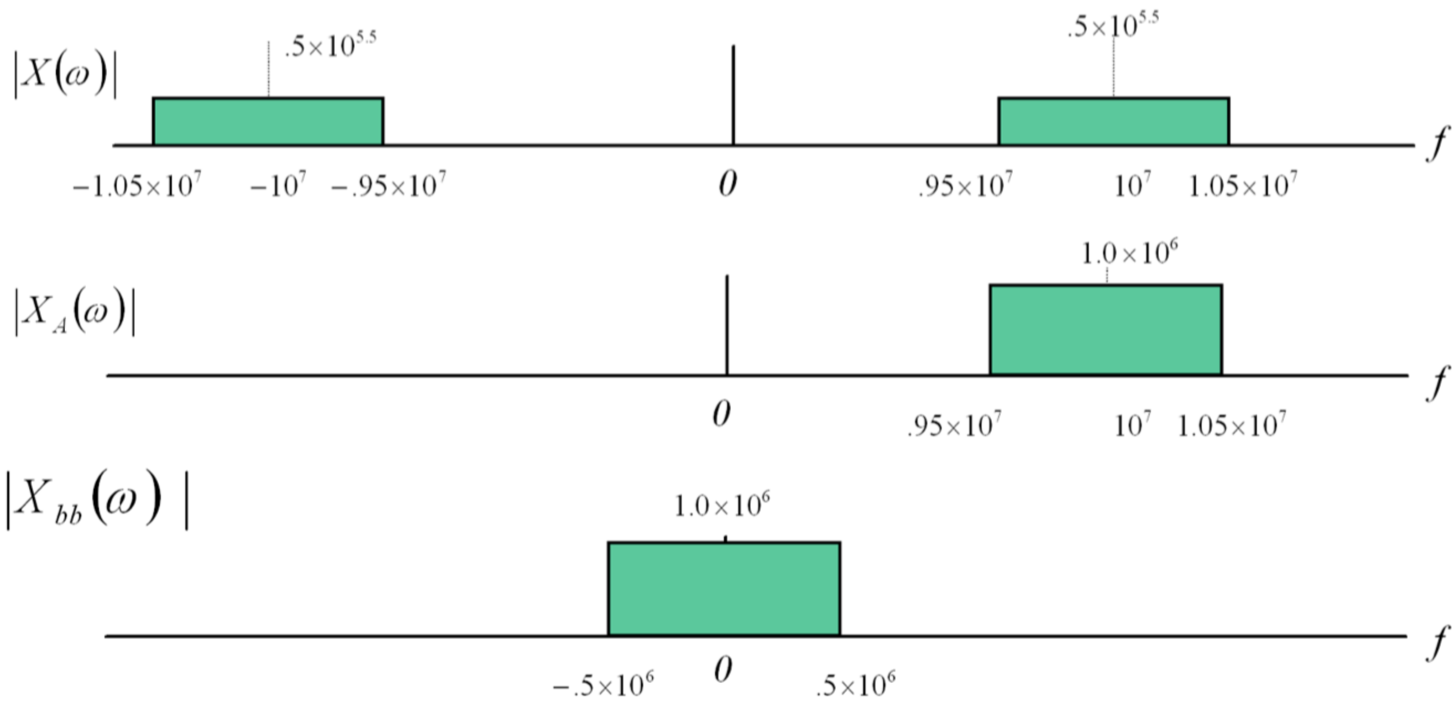
  - $X_{bb}(\omega) = X_A(\omega + \omega_c)$

Baseband spectrum is twice the original spectrum in amplitude, centered around 0.

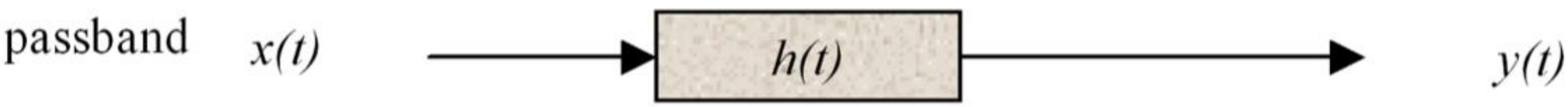
- Reconstruct $X_{bb}(\omega)$ from $X(\omega)$ is possible, and vice versa.

Frequency Spectrum Example

$$x(t) = \text{sinc}(10^6 t) \cdot \cos(2\pi 10^7 t) + 3\text{sinc}(10^6 t) \cdot \sin(2\pi 10^7 t)$$



* Passband Channels

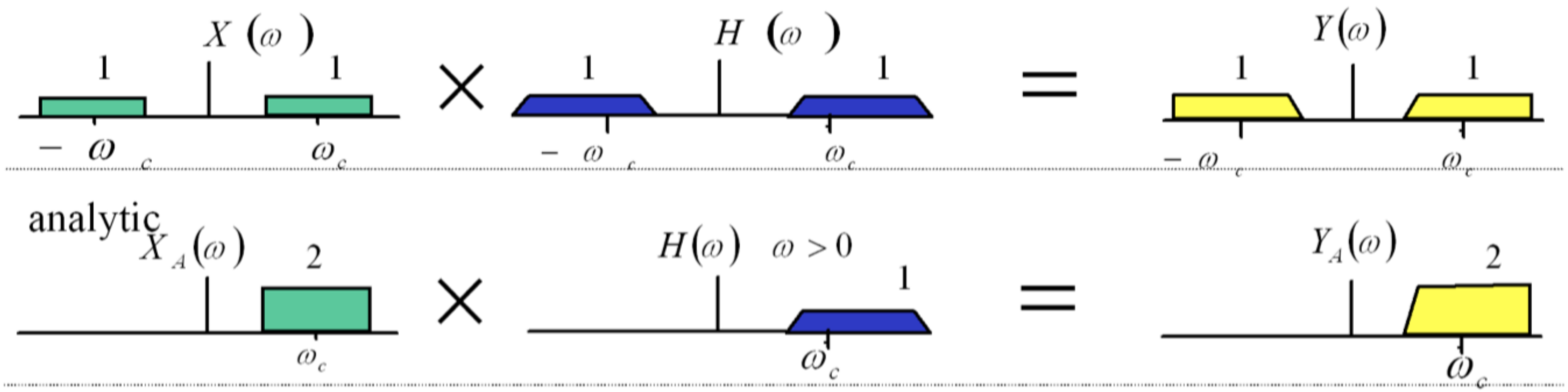


- **Real-valued** channel impulse response (CIR) $h(t)$
- Analytic-equivalent CIR $h_A(t) \triangleq h(t) + j\check{h}(t)$
- Baseband-equivalent CIR *at any carrier frequency* ω_c

$$h_{bb}(t) \triangleq h_A(t)\, \mathrm{e}^{-j\omega_c t}$$

- We know $y(t) = x(t) * h(t)$,
  but what about $y_{bb}(t), x_{bb}(t), h_{bb}(t)$?

* Frequency Domain



$$Y(\omega) = H(\omega) \cdot X(\omega)$$
$$Y_A(\omega) = H(\omega) \cdot X_A(\omega)$$
$$\Rightarrow Y_A(\omega) = \left[H(\omega) \cdot \frac{1}{2}(1 + \mathrm{sgn}(\omega)\right] \cdot X_A(\omega) = \left[\frac{1}{2} H_A(\omega)\right] \cdot X_A(\omega)$$

The factor $\frac{1}{2}(1 + \mathrm{sgn}(\omega))$ extracts the positive component of $H(\omega)$.

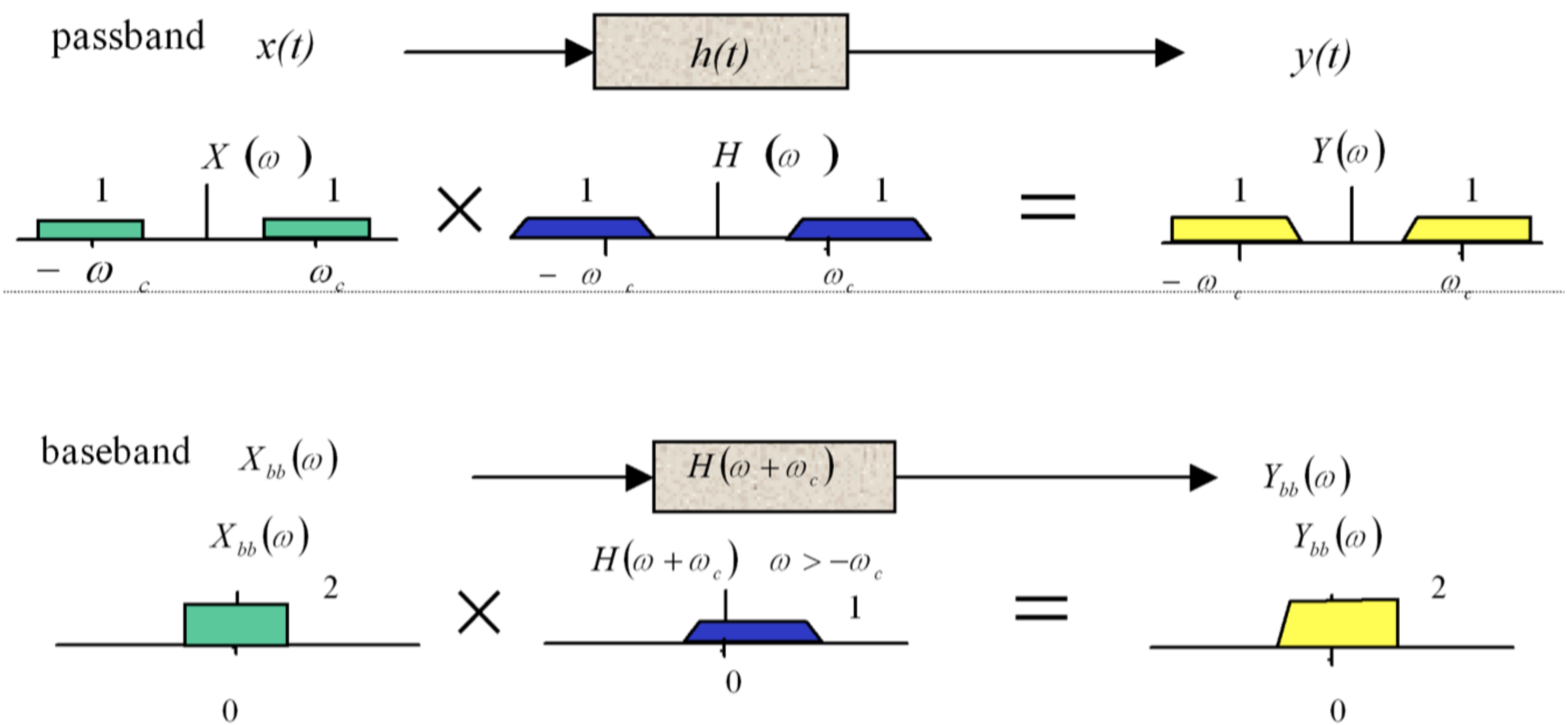
*Baseband Equivalent System

$$Y_{bb}(\omega) = Y_A(\omega + \omega_c) = \left[\frac{1}{2} H_A(\omega + \omega_c)\right] X_A(\omega + \omega_c)$$
$$= \left[\frac{1}{2} H_{bb}(\omega)\right] X_{bb}(\omega)$$
$$= H(\omega + \omega_c) \cdot X_{bb}(\omega)$$

- Passband system $y(t) = x(t) * h(t)$
- Baseband equivalent system
  - $y_{bb}(t) = \frac{1}{2} h_{bb}(t) * x_{bb}(t)$ time domain
  - $Y_{bb}(\omega) = H(\omega + \omega_c) \cdot X_{bb}(\omega)$ frequency domain

The baseband time-domain transfer function is $\frac{1}{2} h_{bb}(t)$, and frequency-domain transfer function is $H(\omega + \omega_c)$.

Baseband Equivalent System



Outline

- Baseband equivalent modeling (no noise)
- Passband modulation schemes
  - Amplitude Shift Keying (ASK)
  - Phase Shift Keying (PSK)
  - Quadrature Amplitude Modulation (QAM)
  - Frequency Shift Keying (FSK)

Amplitude Shift Keying

- Amplitude Shift Keying (ASK)
  - No quadrature component

$$x(t) = \sum_n a_n g(t - nT_s) \cdot \cos(\omega_c t)$$

  - M-ary ASK, with M different amplitude levels

$$a_n \in \{A_0, A_1, \dots, A_{M-1}\}$$

  - Equivalent to baseband M-ary PAM

PAM or ASK?

- Note: in some literature, antipodal constellation scheme is PAM and non-antipodal scheme is ASK.

PAM: $a \in \{-(M-1)A \ldots, -A, A, \ldots, (M-1)A\}$

ASK: $a \in \{0, 2A, \ldots, MA\}$.

- Here we call baseband scheme PAM and passband scheme ASK. Both can have positive or negative symbols.

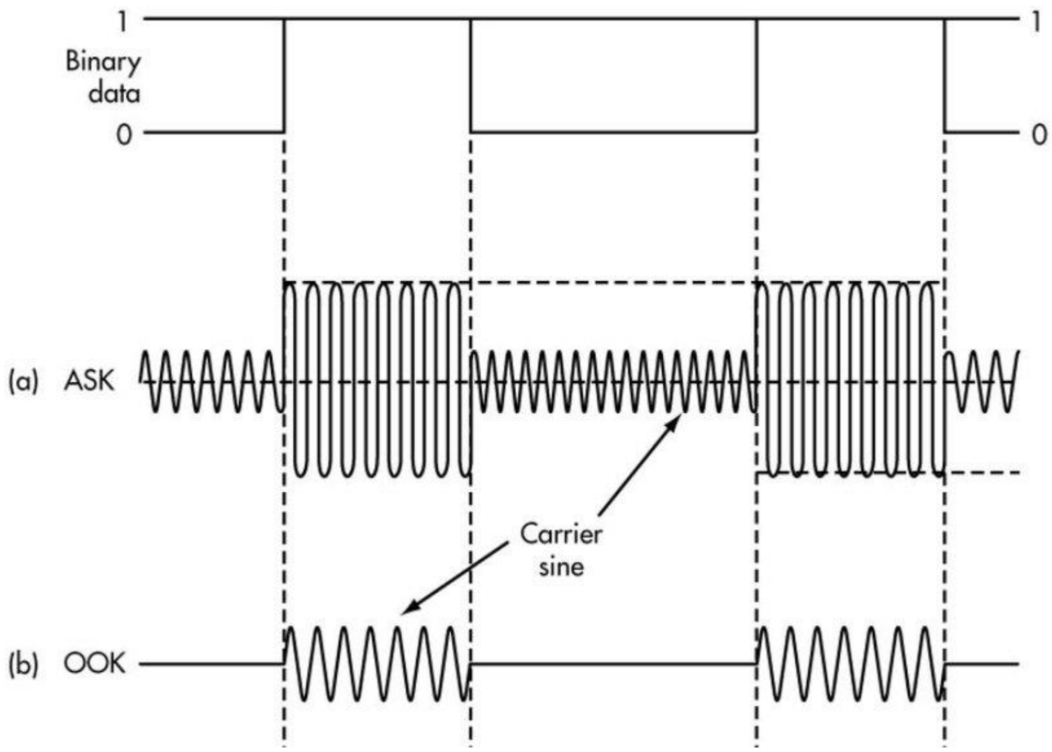
PAM: $x(t) = \sum_n a_n g(t - nT_s)$

ASK: $x(t) = \sum_n a_n g(t - nT_s) \cdot \cos(\omega_c t)$

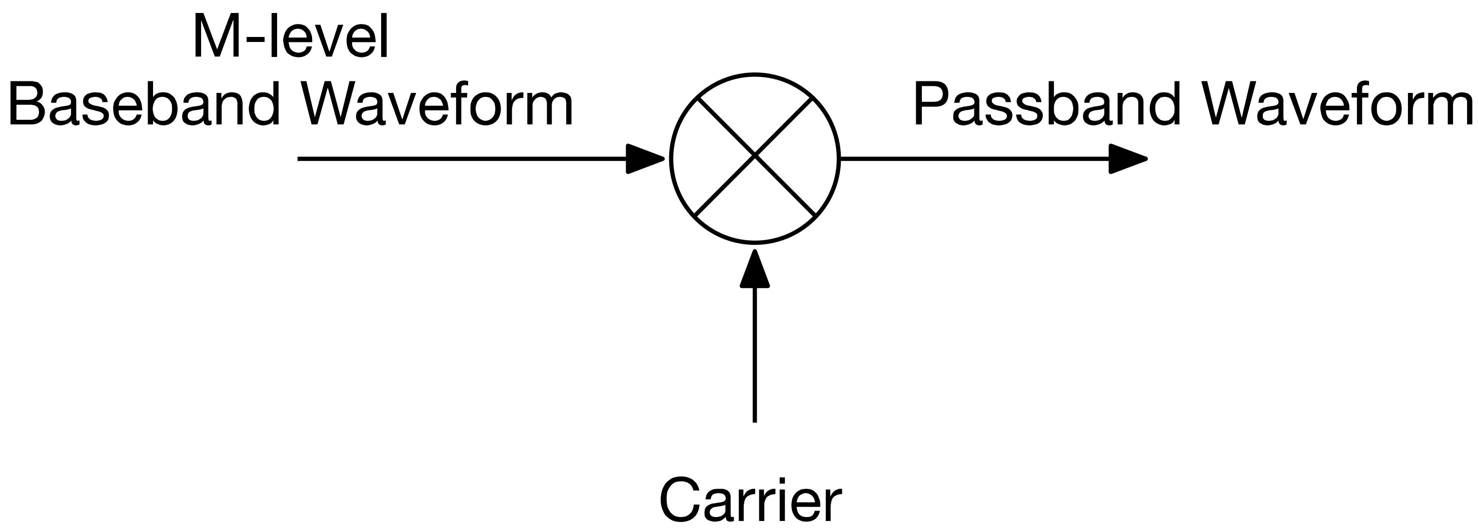
Example: 2ASK

- (a) $a_n \in \{a_1, a_2\}$
- (b) $a_n \in \{0, a\}$

The second can also be called On-Off-Keying (OOK).

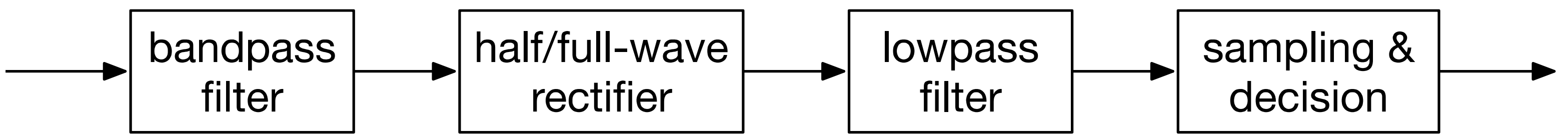


MASK Modulator

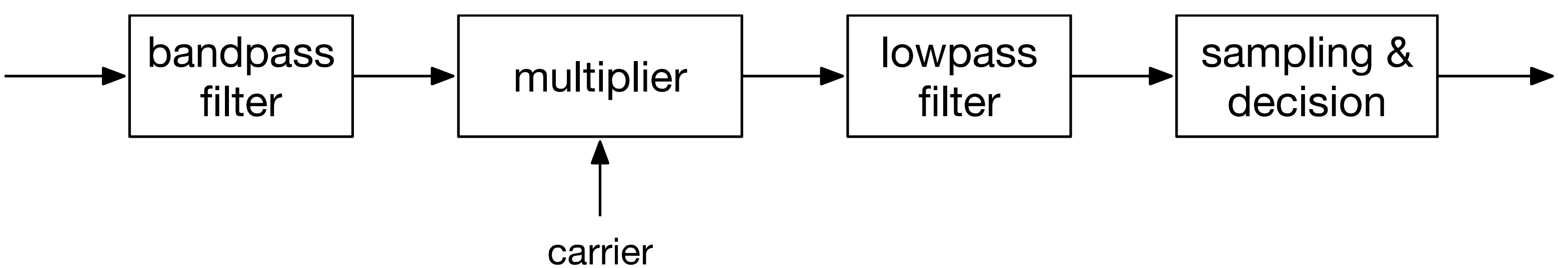


MASK Demodulator

- Envelope Demodulator (non-coherent)
  - Can be used for which kinds of MASK?



- Coherent Demodulator
  - Why does it work?



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Phase Shift Keying

- Phase Shift Keying
  - Constant amplitude, modulate phase

$$x_{MPSK}(t) = \sum_n g(t - nT_s) A \cos(\omega_c t + \phi_n)$$

  - M-ary PSK or MPSK, with M different signals

$$\phi_n = \left\{0, \frac{2\pi}{M}, \frac{4\pi}{M}, \dots, \frac{2(M-1)\pi}{M}\right\}$$

MPSK

- Quadrature decomposition

$$x_{MPSK}(t) = \sum_n g(t - nT_s) A\cos(\omega_c t + \phi_n)$$

$$= \sum_n g(t - nT_s) A(\cos\phi_n \cos(\omega_c t) - \sin\phi_n \sin(\omega_c t))$$

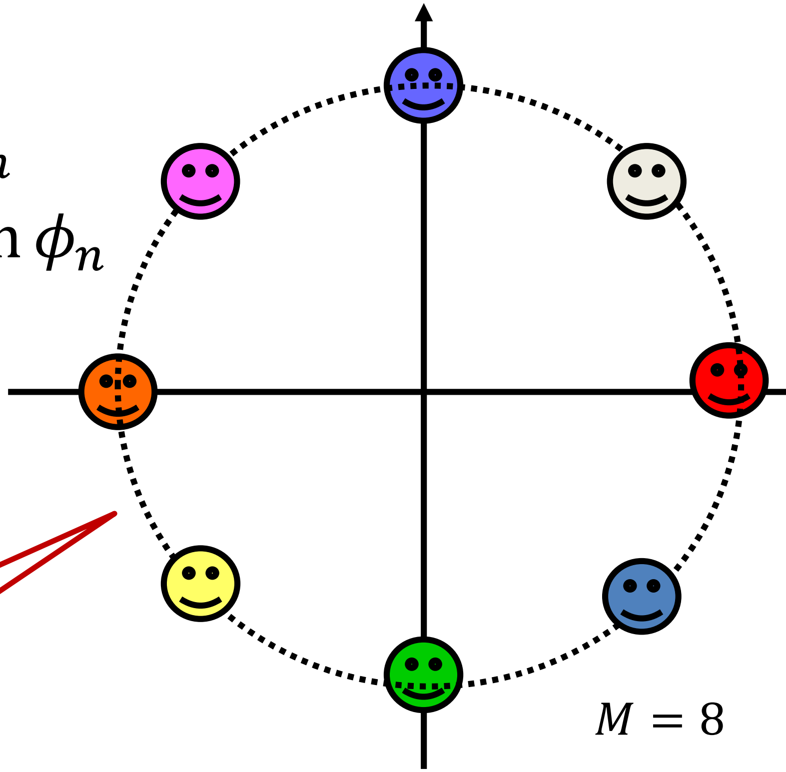
$$= \sum_n [A\cos\phi_n\, g(t - nT_s)] \cos(\omega_c t) +$$

$$\sum_n [A\sin\phi_n\, g(t - nT_s)] (-\sin(\omega_c t))$$

$$\boxed{x_{bb}(t) = \left[\sum_n A\cos\phi_n g(t - nT_s)\right] + j\left[\sum_n A\sin\phi_n g(t - nT_s)\right]}$$

PSK Constellation

- 2-Dimensional symbol
  - Inphase amplitude: $a_I = A\cos\phi_n$
  - Quadrature amplitude: $a_Q = A\sin\phi_n$
- Constellation
  - Complex symbol



$$x_{bb}(t) = \sum_n \boxed{(A\cos\phi_n + jA\sin\phi_n)}\, g(t - nT_s)$$

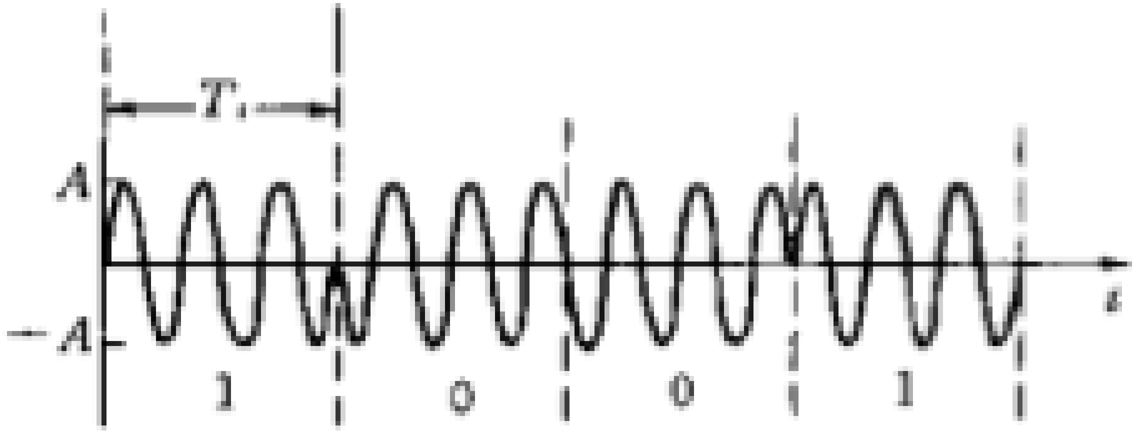
Example—BPSK

- Amplitude

$$a = +A, bit = 0$$

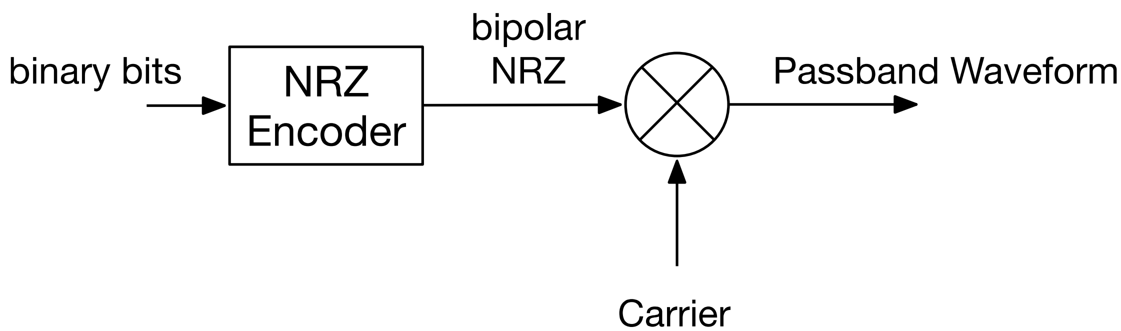
$$a = -A, bit = 1$$

- Waveform

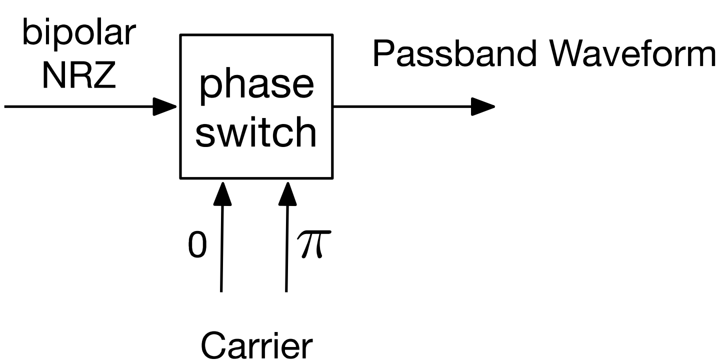


BPSK Modulator

- Multiplier

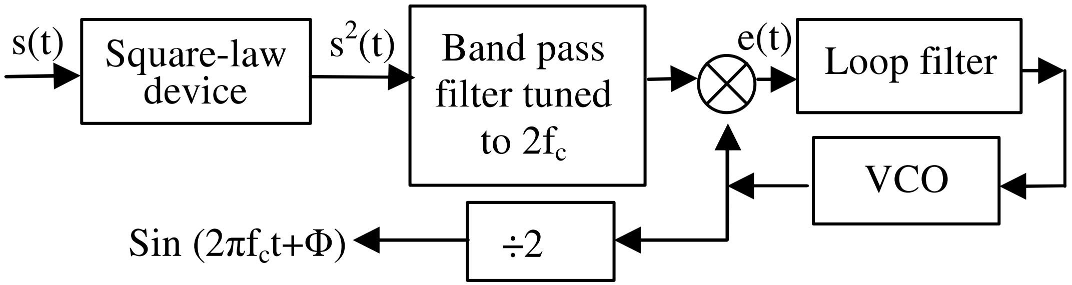


- Phase Selector

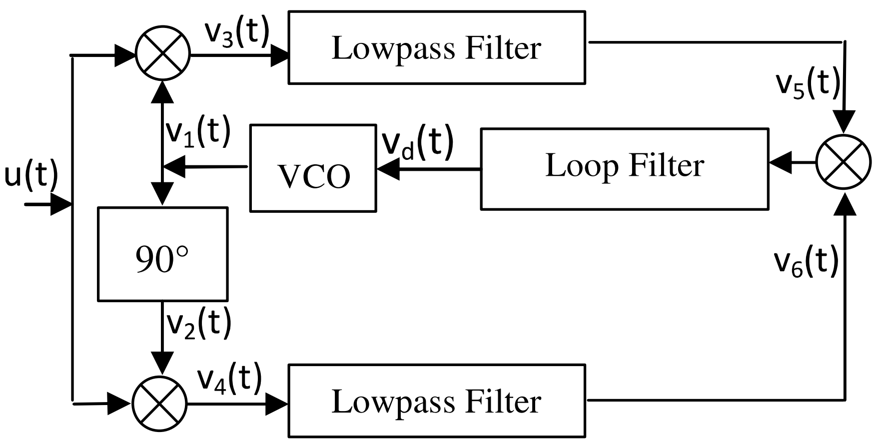


BPSK Demodulator

- Coherent demodulation: how to get carrier?
- Squaring loop



- Costas loop



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Quadrature Amplitude Modulation

- Quadrature Amplitude Modulation
  - 2-dimensional generalization of ASK

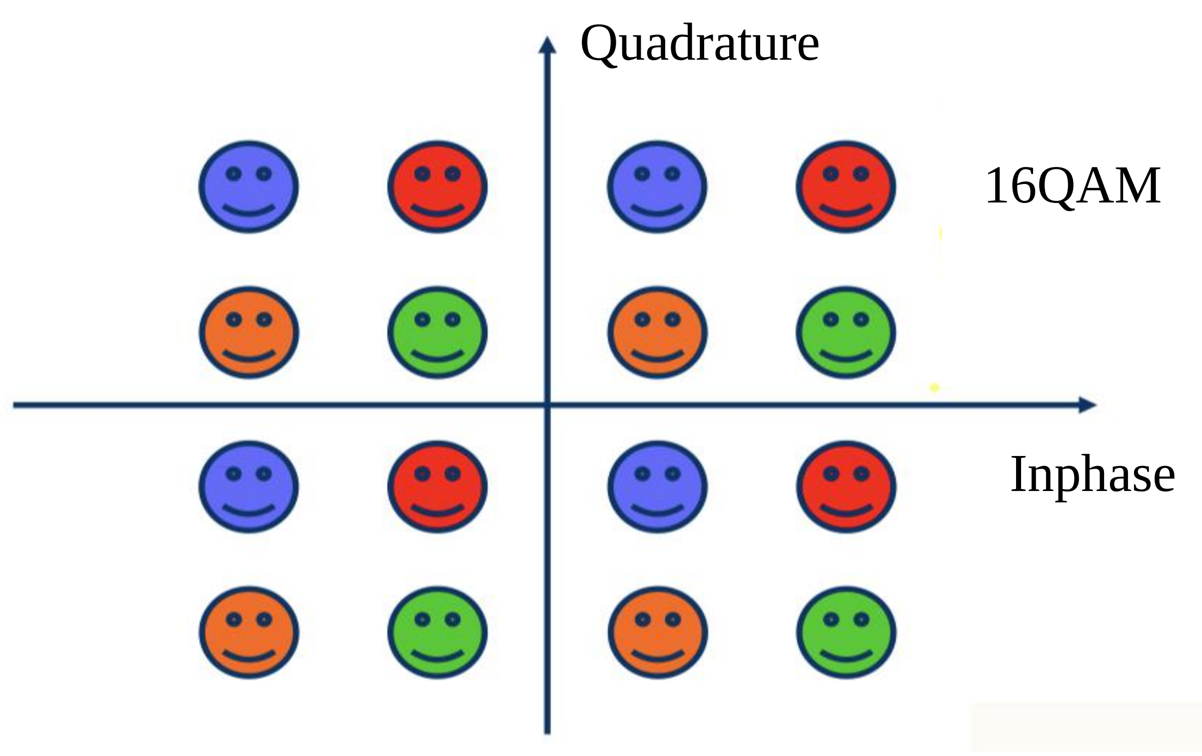
$$x(t) = \left[\sum_n a_n g(t - nT_s)\right] \cos\omega_c t + \left[\sum_n b_n g(t - nT_s)\right](-\sin\omega_c t)$$

$$\boxed{x_{bb}(t) = \left[\sum_n a_n g(t - nT_s)\right] + j\left[\sum_n b_n g(t - nT_s)\right] = \sum_n (a_n + jb_n) g(t - nT_s)}$$

- M-ary QAM constellation
  - Inphase: $a_n \in \{\pm A, \pm 3A, \cdots, \pm(\sqrt{M} - 1)A\}$
  - Quadrature: $b_n \in \{\pm A, \pm 3A, \cdots, \pm(\sqrt{M} - 1)A\}$

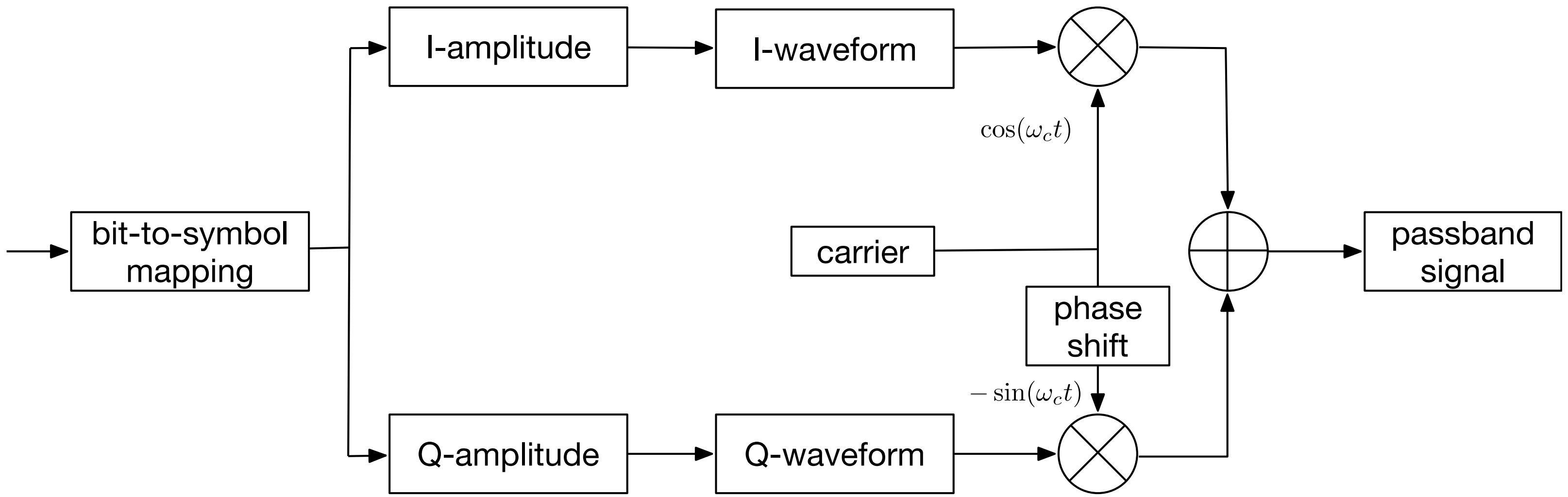
MQAM

- Constellation



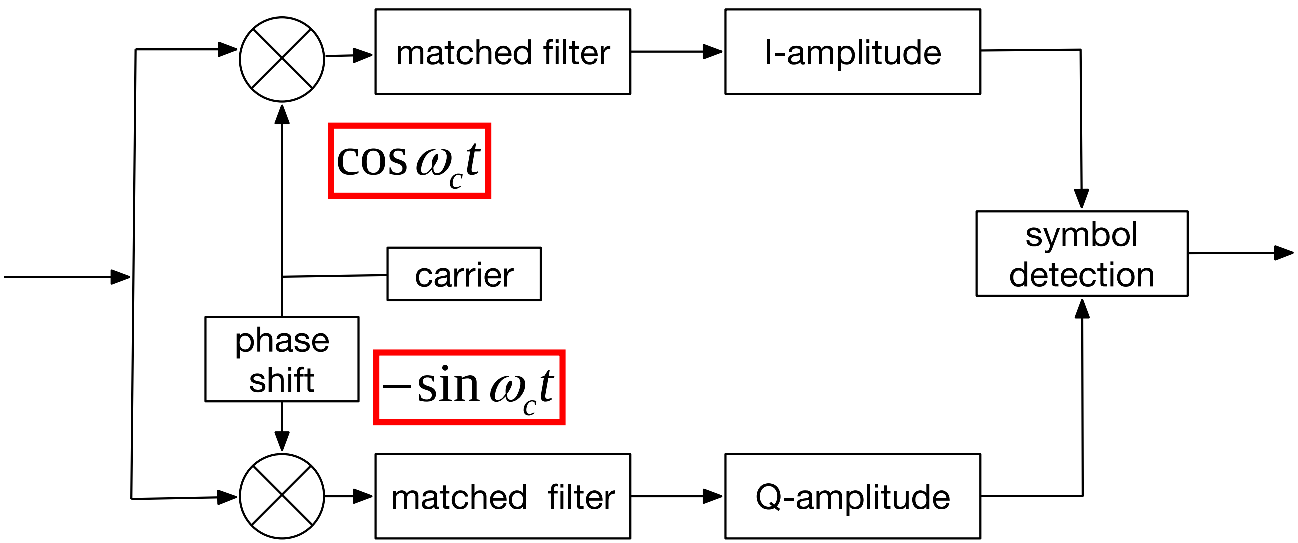
MQAM Modulator

- General I/Q Modulator



MQAM Demodulator

- Coherent Demodulator
  - Can also be used to PSK (WHY?)



MQAM Demodulator

- Coherent Demodulator: Principle
  - Recovering the I-Amplitude

$$\boxed{S(t)\cos\omega_c t = \left[\sum_n a_n g(t-nT_s)\right]\cos\omega_c t\cos\omega_c t + \left[\sum_n b_n g(t-nT_s)\right]\left(-\sin\omega_c t\right)\cos\omega_c t}$$

$$\boxed{\cos\omega_c t\cos\omega_c t = \frac{1}{2}\left[1+\cos 2\omega_c t\right]}$$

$$\boxed{\sin\omega_c t\cos\omega_c t = \frac{1}{2}\sin 2\omega_c t}$$

Removed by LPF/matched Filter

MQAM Demodulator

- Coherent Demodulator: Principle
  - Recovering the Q-Amplitude

$$\boxed{S(t)\sin\omega_c t=\left[\sum_n a_n g(t-nT_s)\right]\cos\omega_c t\sin\omega_c t+\left[\sum_n b_n g(t-nT_s)\right]\left(-\sin\omega_c t\right)\sin\omega_c t}$$

$$\boxed{\sin\omega_c t\sin\omega_c t=\frac{1}{2}[1-\cos 2\omega_c t]}$$

$$\boxed{\sin\omega_c t\cos\omega_c t=\frac{1}{2}\sin 2\omega_c t}$$

Removed by LPF/matched Filter

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Frequency Shift Keying

- Frequency shift keying
  - Different symbols use different carrier frequency.
  - M different signals

$$s_i(t) = \mathrm{A}\cos\big((\omega_c+\omega_i)t\big)$$

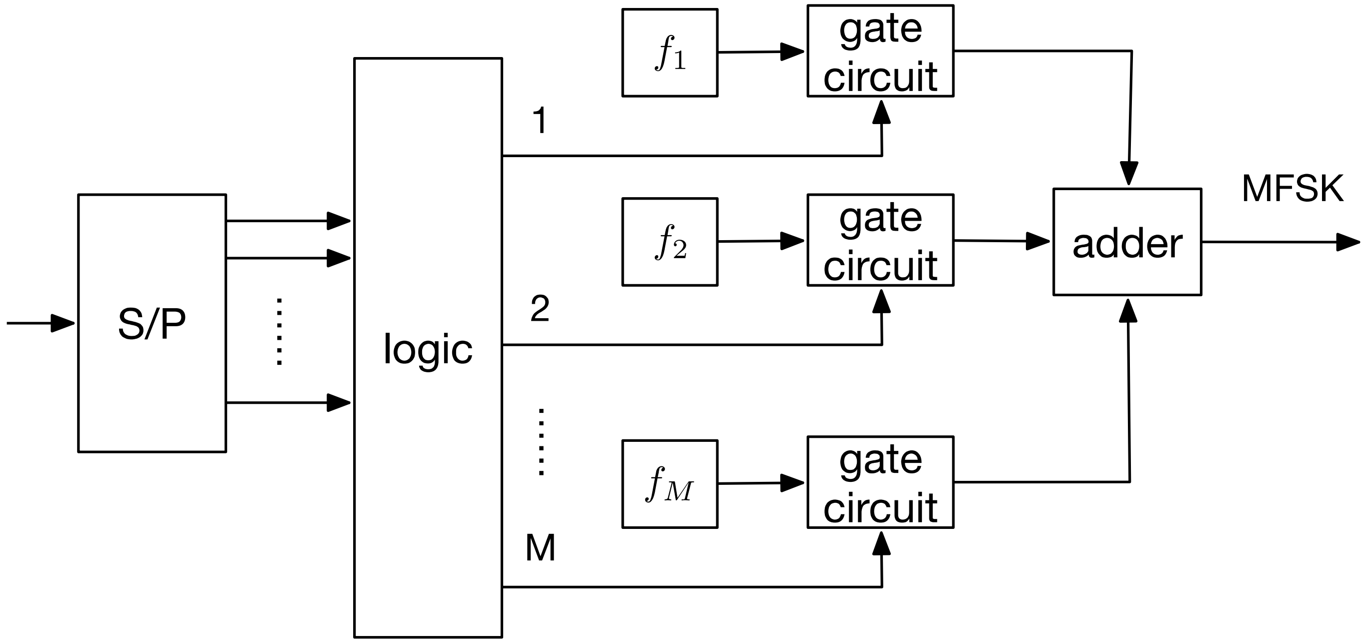
  - Carrier frequencies

$$f_c + f_i = \frac{\omega_c + \omega_i}{2\pi} = \frac{n_c + i}{2T_s}$$

- Signals are designed to be orthogonal

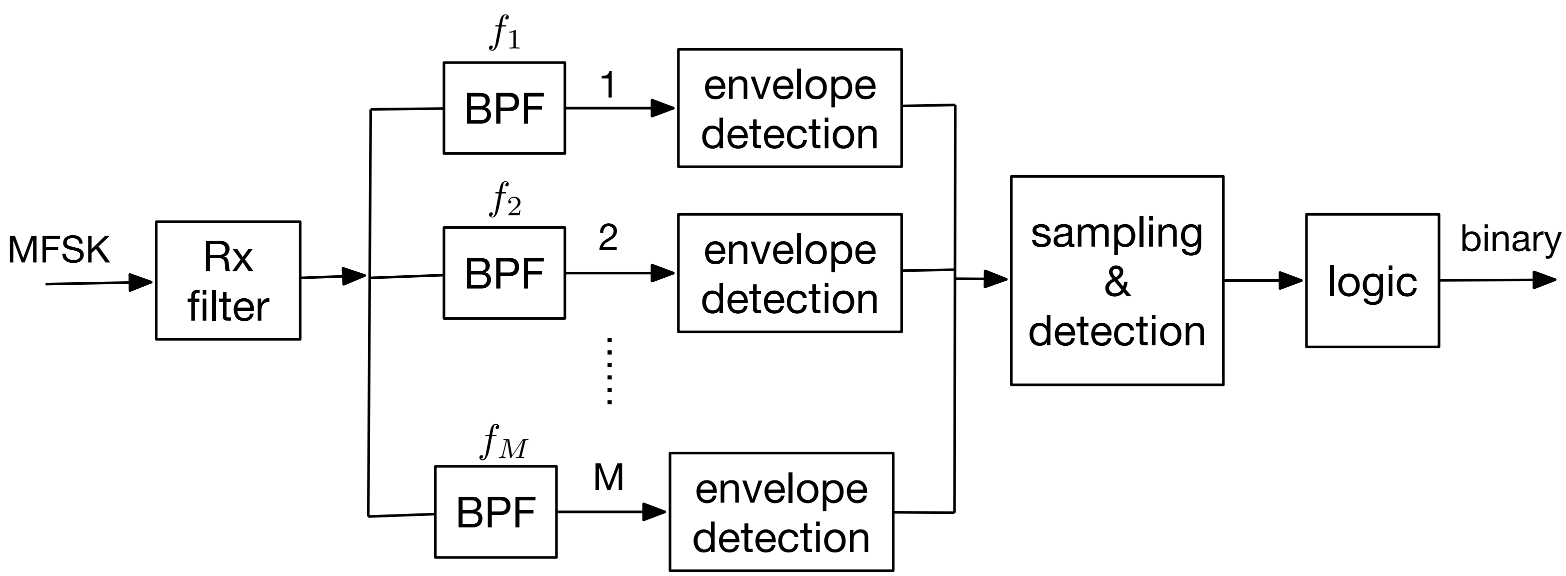
$$\int_0^{T_s} s_i(t)s_j(t)dt = 0, i \neq j$$

MFSK Modulator



MFSK Demodulator

- Non-coherent demodulation



Summary

- Baseband equivalent modeling (Haykin: 6.1, A2.3-A2.4; Proakis: 2.1)
  - Baseband equivalent complex signal
  - Baseband equivalent channel model
- Passband modulation schemes (Haykin: 6.2-6.5; Proakis: 3.2)
  - Amplitude Shift Keying (ASK)
  - Phase Shift Keying (PSK)
  - Quadrature Amplitude Modulation (QAM)
  - Frequency Shift Keying (FSK)